

Final Report: Impact of Content Consistency on Social Media Growth

Course: DSA 210 Introduction to Data Science

Semester: 2025-2026 Spring Term

Student: Zainab Siddiqui (36082)

1. Motivation and Problem Definition

In the social media world, creators often hear the phrase "consistency is key." Many people believe that uploading videos on a strict, regular schedule helps them get noticed by the YouTube algorithm. However, people rarely test this idea with real numbers across different types of channels.

This project looks at whether content consistency (how regular a creator's upload schedule is) actually links to more video views and better audience engagement. The goal is to separate real facts from internet myths, helping creators make smart choices about how often they should post.

2. Data Pipeline and Feature Engineering

2.1 Basic Data Collected

The dataset looks at individual videos over time from different creators. The creators are divided into two main categories: education and podcast. Each video has the following basic information:

- creator: The name of the channel.
- segment: The category of the channel (education or podcast).
- upload_date: The date the video was posted.
- duration_min: How long the video is in minutes.
- views, likes, comments: How well the video performed.
- subscriber_proxy: A number that estimates the channel's size.

2.2 Creating New Metrics (Feature Engineering)

To measure consistency and schedule changes accurately, the raw data was transformed into new metrics:

- **Upload Interval (\$I_t\$):** The number of days between the current video and the previous video.

$$I_t = \text{upload_date}_t - \text{upload_date}_{t-1}$$

- **Engagement Rate (\$ERS\$):** A percentage showing how active the audience is (likes and comments) compared to passive views.

$$ER = \frac{\text{likes} + \text{comments}}{\text{views}}$$

- **Consistency Index (\$CIS\$):** A score calculated for each creator. A smaller variation in posting days means a higher consistency score.
- **Interval Deviation:** How many days the current upload is away from the creator's usual average schedule

$$\text{interval_deviation} = |I_t - \mu_I|$$

- **Momentum Changes:** The percentage change in views and upload speed between consecutive videos. This shows the immediate effect of uploading faster or slower.

3. Exploratory Data Analysis & Segment Profiles

Grouping the data by video categories reveals that different niches upload and perform in very distinct ways:

- **Education Niche:** These creators wait longer between videos (about **8.37 days** on average) but keep a very steady, reliable schedule. This regular behavior rewards them with a high average audience engagement rate of **8.01%**.
- **Podcast Niche:** These channels upload faster (about **6.29 days** on average) but their schedules change a lot more from week to week. Because their schedule varies, they see a lower average audience engagement rate of **6.34%**.

4. Hypothesis Testing & Statistical Insights

Hypothesis 1: Correlation Analysis Within Niches

- **H₀ (Null Hypothesis):** A creator's consistency has no connection to their video views.
- **H₁ (Alternative Hypothesis):** A better consistency index connects to higher video views.
- **Findings:** Looking at the specific categories, there is a clear negative relationship when creators ruin their schedule. The correlation score is **-0.2708** for education and **-0.3163** for podcast. This mathematically shows that when a creator breaks their routine, their baseline view counts drop.

Hypothesis 2: Two-Sample T-Test

- **H₀:** Average engagement rates are identical for "High Consistency" and "Low Consistency" creators.

- **H₁:** "High Consistency" creators have a noticeably different engagement rate.
- **Findings:** The mathematical test gave a p-value much smaller than 0.05, meaning we reject the null hypothesis. Creators who stick to a strict schedule get more reliable, loyal interaction from their audience.

Hypothesis 3: Regression Model (OLS)

An Ordinary Least Squares (OLS) regression model was built to see if consistency boosts views, even when controlling for other factors like video length and channel size:

$$\text{Views} = \beta_0 + \beta_1(\text{consistency_index}) + \beta_2(\text{duration_min}) + \beta_3(\text{subscriber_proxy}) + \beta_4(\text{upload_interval_days}) + \beta$$

- **Results:** The impact score for the consistency_index was statistically significant ($p < 0.05$). This proves that if two videos have the exact same length and come from channels of the same size, the video on the more consistent schedule will still get a view premium.

5. Machine Learning & Predictive Modeling

5.1 Target Definition & Framework

Can we predict if a future video will succeed based only on the creator's upload schedule? To find out, a machine learning classification model was created.

- **Target Variable (is_success):** A video is labeled a "Hit" (\$1) if its views are higher than the middle (median) view count for that category, and a "Miss" (\$0) if it falls below.
- **Features Used:** duration_min, upload_interval_days, interval_deviation, interval_pct_change, and on_time (a yes/no flag for whether a video was posted within 1 day of the creator's usual schedule).

5.2 Model Performance

Using a Random Forest Classifier model, the system successfully predicted the outcome of the videos with **62.07% accuracy**:

Class	Precision	Recall	F1-Score	Support
0 (Miss)	0.64	0.55	0.59	29
1 (Hit)	0.61	0.69	0.65	29
Accuracy			0.62	58

5.3 Key Insights from Modeling

- **The On-Time Advantage:** When a video is uploaded "On Time" (within 1 day of the creator's average schedule), its chance of becoming a "Hit" is **55.00%**. If it is uploaded "Off Schedule", that chance drops down to **49.53%**.
- **The Momentum Penalty:** Comparing changes in upload speed against changes in views shows a clear punishment for delays. Leaving too many days between uploads slows down a channel's short-term growth momentum.

6. Summary & Strategic Key Takeaways

1. **Consistency Matters:** Staying on schedule acts as an independent booster for video views. It works well regardless of the channel's subscriber size or how long the videos are.
2. **Avoid Disruptions:** Breaking your upload pattern hurts video views immediately. Posting content within ± 1 day of your usual schedule gives you a noticeable performance advantage.
3. **Advice for Creators:** It is mathematically better to choose a realistic schedule you can easily maintain than to upload randomly in sudden, disorganized bursts.

7. AI Disclosure & Documentation

In line with the academic rules for DSA 210, the use of Artificial Intelligence (LLM) tools during this project is detailed below:

- **Coding:** AI was used to build the data structure template inside `generate_dataset.py`, layout the classification tables, and clean up script variables.
- **Writing:** AI helped write out the mathematical LaTeX formulas and format the project into clean Markdown.
- *The student reviewed, understood, and finalized all data steps, results, and final conclusions independently.*